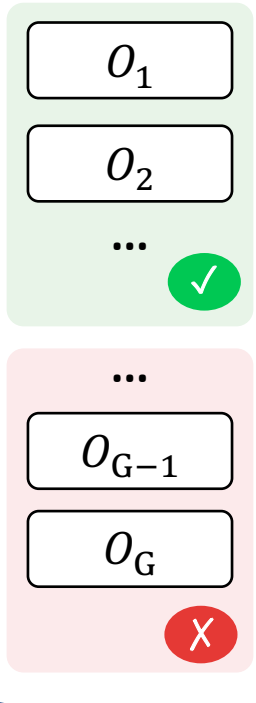


Update Policy

Model Outputs

Policy model π_θ

Question x



Student Mode

$$\pi_\theta(\cdot | x, y)$$

Teacher Mode

$$\pi_\theta(\cdot | x, r, y)$$

OPSD's Core Practice

Privileged r

RLSD's Practice

$$\Delta_t = \text{sg}(\log P_T(y_t) - \log P_S(y_t))$$

$$w_t = \exp(\text{sign}(A) \cdot \Delta_t) = \left(\frac{P_T}{P_S}\right)^{\text{sign}(A)}$$

Token-Level Dense Signal

Correctness from environment

Verifier

$$R(x, y)$$

Advantage

$$A = (R - \mu) / \sigma$$

GRPO's Core Practice

Control the magnitude

$$w_1^1, w_1^2, \dots, w_1^{|y|}$$

$$w_2^1, w_2^2, \dots, w_2^{|y|}$$

$$w_n^1, w_n^2, \dots, w_n^{|y|}$$

$$w_G^1, w_G^2, \dots, w_G^{|y|}$$

$$P_T / P_S$$

Teacher supports

Teacher disfavors

Control the direction

$$A_1$$

$$A_2$$

$$A_n$$

$$A_G$$

Token-Level Advantages

$$\hat{A}_1$$

$$\hat{A}_2$$

$$\hat{A}_n$$

$$\hat{A}_G$$

RLSD's Practice